

A Multi-Modal Ensemble Learning Approach for Early Survival Prediction

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INTRODUCTION

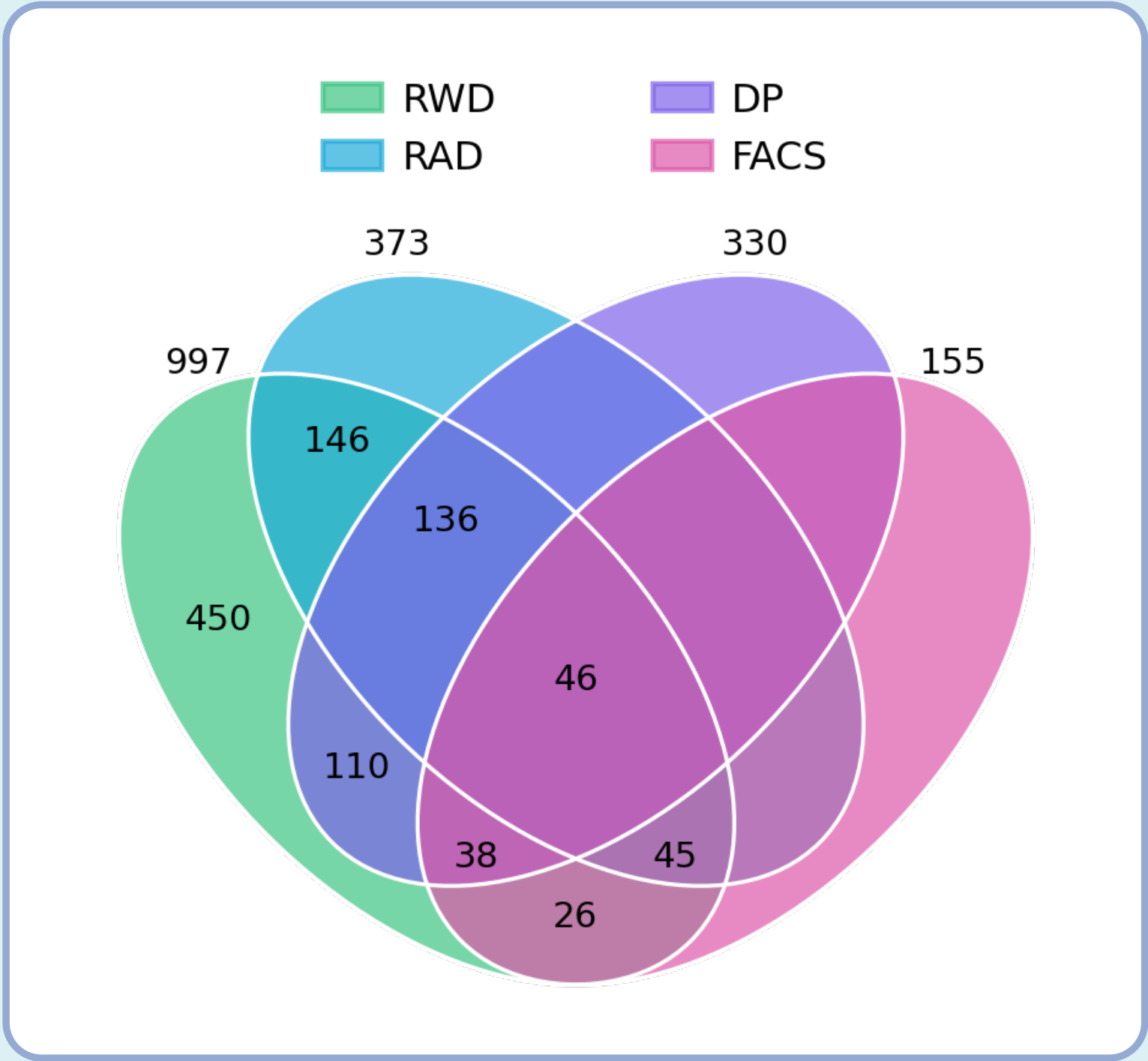
- Early prediction of **6-month survival** in lung cancer patients is critical for tailoring treatment decisions and improving outcomes
- Modern diagnostics provide **multi-modal data**, including clinical, radiological, digital pathology and molecular features
- However, in **real-world clinical settings**, many patients have **missing modalities** due to practical constraints
- This poses a major challenge for **traditional machine learning models**, which typically rely on complete data from all sources

GOALS

- Develop a **model** that utilizes **all** available patient **data**
- Improve predictions by **integrating multi-modal information**
- Ensure **robustness to missing modalities**
- Preserve **model interpretability** despite data complexity

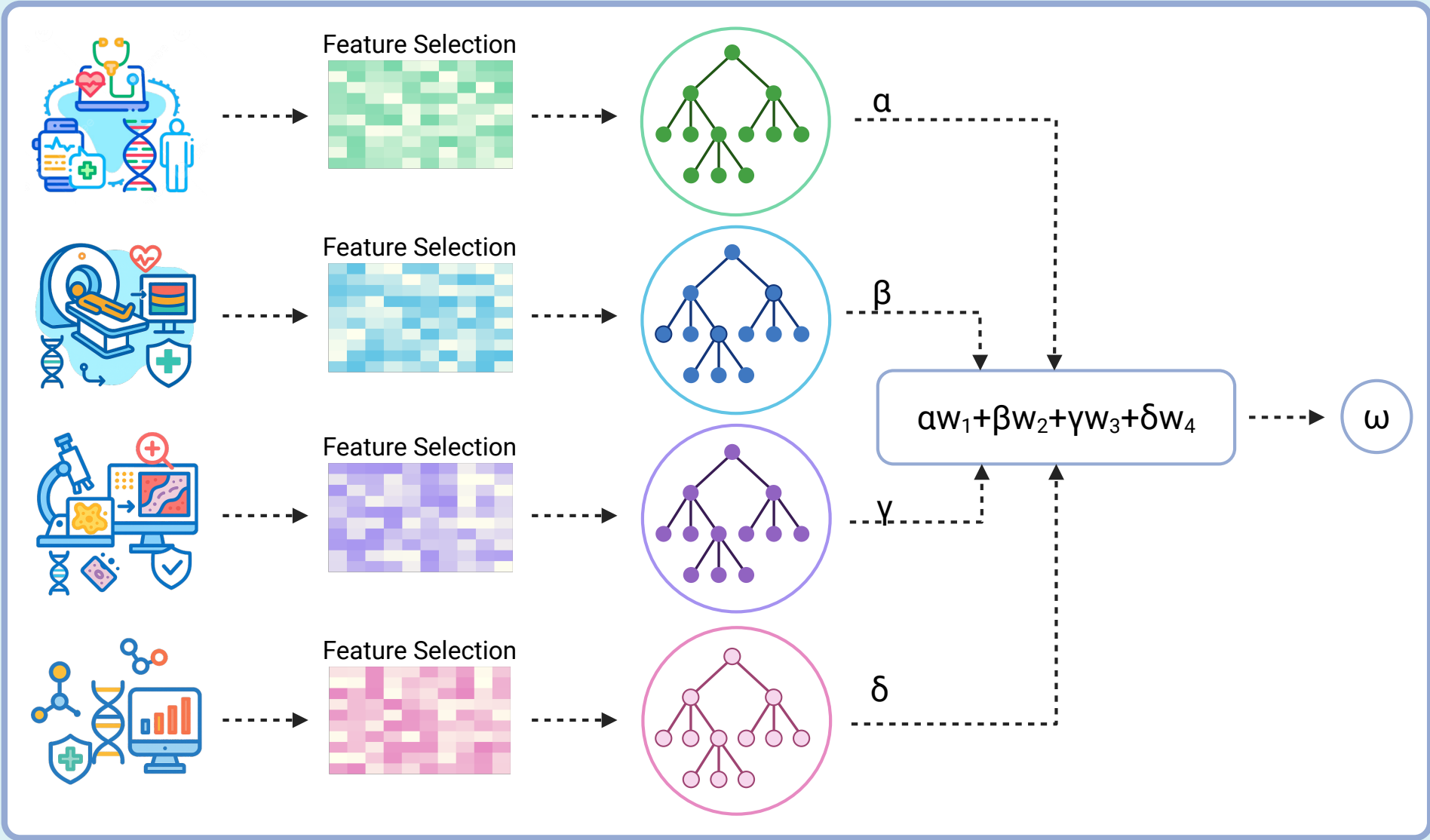
DATASET

- **APOLLO 11** is a 3-year Italian project involving **48 centers** focused on lung cancer
- Builds a **collaborative network and biobank** for data collection
- Aims to develop an **AI-based biomarker** for predicting immunotherapy response in **NSCLC** patients
- Includes Clinical Data (**RWD**), Radiomics (**RAD**), Digital Pathology (**DP**) and Fluorescence-activated Cell Sorting (**FACS**)
- Dataset split into **training (90%)** and **test (10%)** subsets.
- **Test set distribution** matches the training set across all modalities



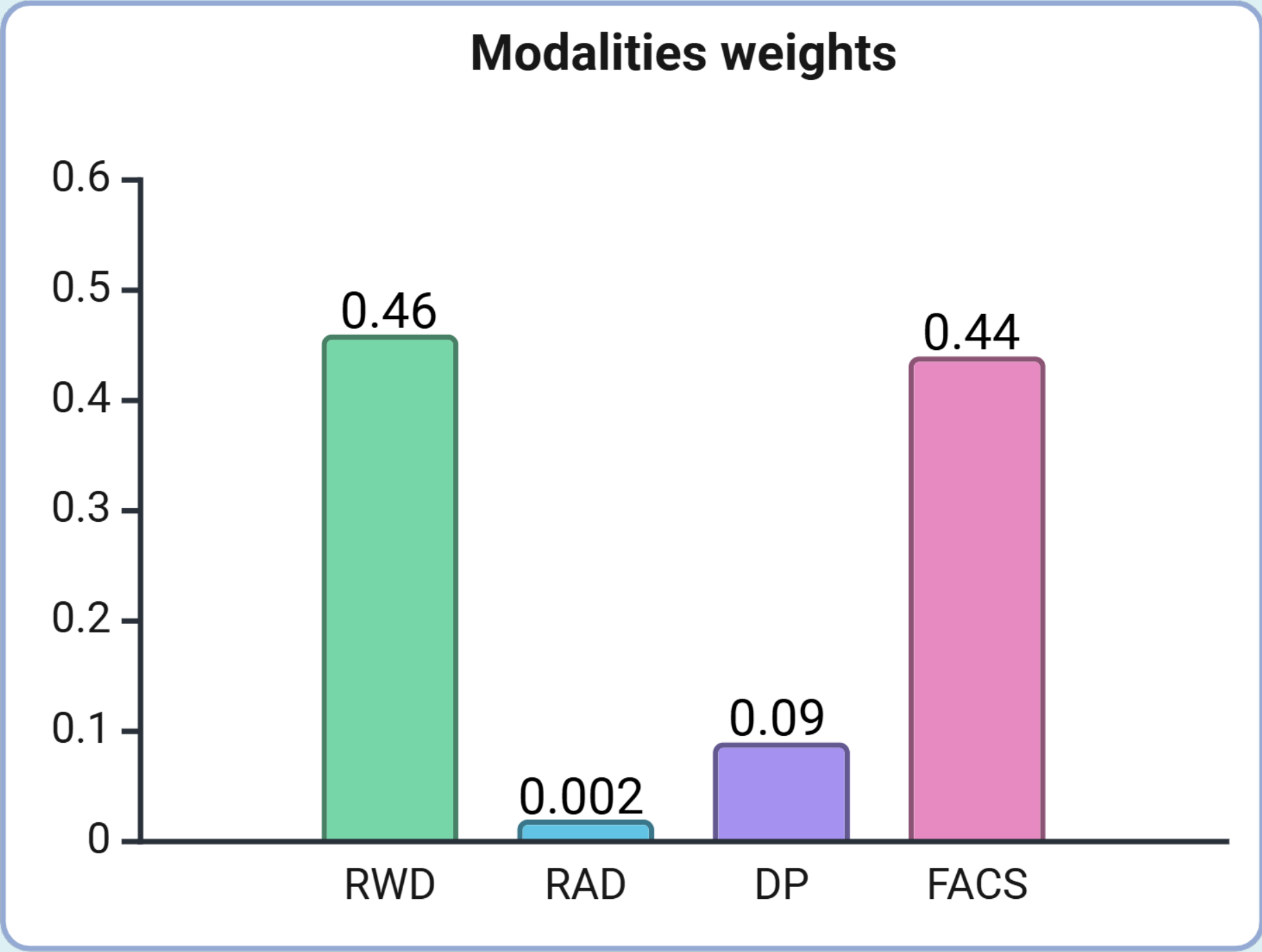
METHODS

- **Modality-specific models** were trained independently
- **Logistic Regression** for RWD and FACS, **Random Forest** for Radiomics and Digital Pathology
- **Ensemble model** combines outputs via a **weighted sum**
- Weights are optimized through **5-fold cross-validation**



RESULTS

- **Evaluation metrics:** F1-macro, AUC, and accuracy
- **Comparison:** Multimodal model vs. best unimodal baseline (RWD-only)
- **Outcome:** All metrics improved → **benefit of multimodal integration confirmed**
- **Robustness:** Maintains performance with missing modalities → **strong generalizability**
- **Modality importance:** Only **RWD** and **FACS** had significant weight → indicates dominant predictive power
- **Stability:** **Lower standard deviation** in the multimodal model → **greater reliability** and **clinical consistency**



Cross-validation scores		
Metric	Unimodal RWD	Multimodal
F1	0.678 ± 0.052	0.680 ± 0.050
AUC	0.699 ± 0.058	0.702 ± 0.055
ACCURACY	0.698 ± 0.049	0.700 ± 0.046
Train scores		
Metric	Unimodal RWD	Multimodal
F1	0.701	0.713
AUC	0.725	0.737
ACCURACY	0.718	0.730
Test scores		
Metric	Unimodal RWD	Multimodal
F1	0.635	0.648
AUC	0.685	0.687
ACCURACY	0.660	0.680

CONCLUSIONS

- Developed a **multimodal AI model** to predict 6-month survival in NSCLC patients
- **Outperformed** all **single-modality** baselines, particularly RWD-only
- Demonstrated **robustness to missing data**, enhancing real-world applicability
- Preserved **interpretability**, with clear attribution of modality contributions
- Findings support the **integration of heterogeneous clinical data** to improve prediction in lung cancer care

Next steps:

- **Expand the dataset** to include more diverse patient populations
- **Incorporate additional modalities** to further enhance model richness
- **Perform external validation** on independent cohorts to confirm generalizability and robustness